QUANT

Averages, LCM & Decimals

Assignment 05

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1. In a class with a certain number of students if one student weighing 50 kg is added then the average weight of the class increases by 1 kg. If one more student weighing 50 kg is added then the average weight of the class increases by 1.5 kg over the original average. What is the original average weight (in kg) of the class?

1. 46 2. 4 3. 2 4. 47

2. Find the average of 11, 22, 33, 44,....., 1331.

1. 561 2. 781 3. 671 4. 572

3. The average of nine consecutive numbers is n. If the next two numbers are also included the new average will

1. increase by 2 2. remain the same 3. increase by 1.5 4. increase by 1

4. The average score of a class of boys and girls in an examination is A. The ratio of boys and girls in the class is 3: 1. If the average score of the boys is (A + 1), the average score of the girls is:

1. (A-1) 2. (A-3) 3. (A+1) 4. (A+3)

5. The average of three numbers is 77. If the first number is twice that of second number and the second number is twice that of third number, then the first number is

1. 132 2. 66 3. 33 4. 99

6. Average age of employees in an organization is 40 years. When 120 new employees whose average age is 32 years joined the organization, the average age is decreased by 4 years. Find the number of employees of the organization after joining of the new employees.

1. 1200 2. 120 3. 360 4. 240

7. The sum of five numbers is 555. The average of first two numbers is 75 and the third number is 115. What is the average of last two numbers?

1. 145 2. 150 3. 265 4. 290

8. Of three numbers, the second is thrice the first and the third number is three-fourth of the first. If the average of the three numbers is 114, the largest number is

1. 72 2. 216 3. 354 4. 726

9. The average of two numbers A and B is 20, that of B and C is 19 and of C and A it is 21. What is the value of A?

1. 24 2. 22 3. 20 4. 18

10. The ratio of ages of a husband and wife when they got married was 6:5. They had two children 4 years and 6 years after they got married. The sum of present ages of husband and wife is 6.4 times the sum of the ages of their children. The average age of family at present is 18.5 years. Find the ratio of ages of husband and wife when their second child was born.

1. 7 :6 2. 15 :13 3. 6 :7 4. None of these

Answer Key

Test - 05	
1.	4
2.	3
3.	4
4.	2
5.	1
6.	4
7.	1
8.	2
9.	2
10.	2

Test - 05

1. Let x be the original average and n be the number of students,
From the first increase in the average, we get nx + 50 = (n + 1) (x + 1) => 50 – x = n + 1.....(1)
From the second increase in the average, we get nx + 100 = (n + 2)(x + 1.5) =>100 – 2x = 1.5(n+2).....(2)
Solving (1) and (2), we get the value of x = 47.

2. The given numbers are in A.P. If numbers are in A.P. and if first and last number is given then the sum of numbers is
n (a + l)/2 and the average is (a + l)/ 2. So the average of given numbers is (11 + 1331)/ 2 = 671.

3. As mentioned here, consecutive numbers, so the difference between any two consecutive numbers will be 1. The average of 1st 9 no. is 5th term (behind the middle term) $\Rightarrow T_5 = n \Rightarrow T_6 = n+1$
And the average of 1st 11 terms will be 6th term (middle term) So, Average will increase by 1.

4. Let the no. of boys =3x ∴no. of girls =x

Now average $= \dfrac{score\,of\,boys + score\,of\,girls}{Total\,no.\,of\,students}$

$\therefore A = \dfrac{3x(A+1)+x(y)}{4x}$

$\Rightarrow 4A = 3(A + 1) + y \Rightarrow 4A = 3A + 3 + y$
$\Rightarrow y = A - 3$. Hence answer is 2nd option

5. Let the three numbers be a,b and c

So, $\dfrac{a+b+c}{3} = 77 \Rightarrow a + b + c = 231$ …….(1)

Now, we are given that $a = 2b \Rightarrow b = \dfrac{a}{2}$

And $b = 2c \Rightarrow c = \dfrac{b}{2} = \dfrac{a}{4}$

Using equation

(1), $a + \dfrac{a}{2} + \dfrac{a}{4} = 231 \Rightarrow \dfrac{7a}{4} = 231 \Rightarrow a = 132$

So, Answer is option 1

6. Let the original employees = x.
Now 40x + 120 × 32 = (x + 120) × 36. On solving we get x = 120. So the total employees = 120 + 120 = 240.

7. a + b + c + d + e = 555 –(1)
Average of first two numbers is 75
Therefore a + b = 2 × 75 = 150
The third number is given to us as 115
Therefore a + b + c = 115 + 150 = 265
Using it with 1st equation, d + e = 290
Therefore the average of two Numbers is 290/2 = 145.

8. Let the first no be 'a' so, the second and third number will be 3a and $\dfrac{3a}{4}$ respectively

Now applying given conditions, =

$\dfrac{a + 3a + \dfrac{3a}{4}}{3} = 114 \Rightarrow a = 72$. So, second and third number will be 3×72, i.e. 216 and 54 respectively.
Thus the largest number will be 216.

9. According to the given statements, we get
A + B = 40, B + C = 38, C +A = 42
On adding,
2(A + B + C) = 40 + 38 + 42 = 120
⇒ A + B + C = 60
Therefore, A = (A+ B +C) – (B + C) = 60 – 38 = 22

10. Total present age of family = 18.5 × 4 = 74 years.
As ratio of (husband +wife) : children is 6.4 : 1 means it is 64 years and 10 years respectively.
As difference of ages of two children is 2 years, so they must be of 6 years and 4 years respectively.
So, the marriage of their parents must have taken place 6 + 4 i.e. 10 years back and at that time their sum of ages was 64 – 20 = 44 years.
Also, ratio of their ages at that time is 6 : 5. So, husband's age at the time of marriage was 24 years and that of wife was 20 years.
So, the required ratio when the second child is born is 30 : 26 i.e. 15 : 13.